

Predicate Abstraction

Viktor Kunčák

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Abstract interpretation domain is determined by a set of formulas (predicates) $\mathcal{P}$ on program variables.

Example: $\mathcal{P} = \{P_0, P_1, P_2, P_3\}$ where

$$P_0 \equiv \text{false}$$

$$P_1 \equiv 0 < x$$

$$P_2 \equiv 0 < y$$

$$P_3 \equiv x < y$$

Analysis tries to construct invariants from these predicates using

- ▶ conjunctions, e.g. $P_1 \wedge P_3$
- ▶ more generally, conjunctions and disjunctions, e.g. $P_3 \wedge (P_1 \vee P_2)$

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For now: we consider only conjunctions.

We assume $P_0 \equiv \text{false}$, other predicates in $\mathcal{P}$ are arbitrary

- ▶ expressed in a logic for which we have a theorem prover

Example of Analysis Result

$$\mathcal{P} = \{false, 0 < x, 0 \leq x, 0 < y, x < y, x = 0, y = 1, x < 1000, 1000 \leq x\}$$

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x = 0;
y = 1;
// 0 < y, x < y, x = 0, y = 1, x < 1000
// 0 < y, 0 ≤ x, x < y
while (x < 1000) {
    // 0 < y, 0 ≤ x, x < y, x < 1000
    x = x + 1;
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    y = 2 * x;
    // 0 < y, 0 ≤ x, 0 < x, x < y
    y = y + 1;
    // 0 < y, 0 ≤ x, 0 < x, x < y
    print(y);
}
// 0 < y, 0 ≤ x, x < y, 1000 ≤ x
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Lattice of Conjunctions of Predicates and Concretization

$\mathcal{P} = \{P_0, P_1, \dots, P_n\}$ - predicates

► formulas whose free variables denote program variables

$A = 2^{\mathcal{P}}$, so for $a \in A$ we have $a \subseteq \mathcal{P}$

Example: $a_0 = \{0 < x, x < y\}$.

$s \models F$ means: formula F is true for variables given by the program state s

$$\gamma(a) = \{s \mid s \models \bigwedge_{P \in a} P\}$$

Shorthand: $\bigwedge a$ means $\bigwedge_{P \in a} P$

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Define:

$$a_1 \sqsubseteq a_2 \iff a_2 \subseteq a_1$$

Lemma: $a_1 \sqsubseteq a_2 \rightarrow \gamma(a_1) \subseteq \gamma(a_2)$

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Lemma: $a_1 \sqsubseteq a_2 \rightarrow \gamma(a_1) \subseteq \gamma(a_2)$

Does the converse hold?

Size of the Lattice

$$\{\text{false}, 0 < x, x < y\} \sqsubseteq \{0 < x, 0 < y\} \sqsubseteq \{0 < x\} \sqsubseteq \emptyset$$

Draw the Hasse diagram for the lattice $(A, \sqsubseteq)$ i.e. $(2^{\mathcal{P}}, \supseteq)$ for $\mathcal{P} = \{P_0, P_1, P_2\}$ a three-element set.

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Do $\sqcup$ and $\sqcap$ exist?

Galois Connection

For $\gamma(a) = \{s \mid s \models \bigwedge_{P \in a} P\}$ we define

$$\alpha(c) = \{P \in \mathcal{P} \mid \forall s \in c. s \models P\}$$

$$\alpha(\{(-1, 1)\}) =$$

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Is (α, γ) a Galois connection between $(A, \sqsubseteq)$ and $(C, \subseteq)$?

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Shorthand: in logic, if M is a set of assignments to variables (structures) and $\mathcal{A}$ is a set of formulas (e.g. axioms), then $M \models \mathcal{A}$ means

$$\forall m \in M. \forall F \in \mathcal{A}. m \models F$$

So, both conditions of Galois connection reduce to $c \models a$

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We show this is not the case. This is because γ is not injective.

Indeed, take $a_1 = \{false\}$ and $a_2 = \{false, x > 0\}$. Then

$$\gamma(a_1) = \emptyset = \gamma(a_2)$$

Note $\alpha(\gamma(a_1)) = \alpha(\mathcal{P}) = \alpha(\gamma(a_2))$, but $a_1 \neq a_2$, but it is not the case that $a_1 = a_2$. In this particular case,

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Can you find an example of non-injectivity in our 4 predicates that does not involve false?

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Therefore $P \in \alpha(c_1)$. We showed $c_2 \subseteq c_1$, so $c_1 \sqsubseteq c_2$.

Computing Approximate Strongest Postcondition

$$\mathcal{P} = \{false, 0 < x, 0 < y, x < y\}$$

Consider computing $sp^\#(\{0 < x\}, y := x + 1)$. We can test for each predicate $P' \in \mathcal{P}$ whether

$$x > 0 \wedge (y' = x + 1 \wedge x' = x) \implies P'(x', y')$$

We obtain that the condition holds for $0 < x$, $0 < y$, and for $x < y$, but not for *false*. Thus,

$$sp^\#(\{0 < x\}, y := x + 1) = \{0 < x, 0 < y, x < y\}$$

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$$sp^\#(\{0 < x, x < y\}, x := x - 1) = \{0 < y, x < y\}$$

What is the relation between $\{0 < x, x < y\}$ and $\{0 < x, 0 < y, x < y\}$?

Deriving Rule for Computing sp

Fix some command given by relation r .

Denote $a' = sp^\#(a, r)$. We are computing a' . For correctness we need

$$sp(\gamma(a), r) \subseteq \gamma(a')$$

Thanks to Galois connection, this is equivalent to

$$\alpha(sp(\gamma(a), r)) \sqsubseteq a'$$

We wish to find the smallest lattice element a' , which is the largest set (this gives the tightest approximation). So we let

$$a' = \alpha(sp(\gamma(a), r))$$

Given that $\gamma(a) = \{s \mid s \models \bigwedge a\}$, and $\alpha(c) = \{P \in \mathcal{P} \mid \forall s \in c. s \models P\}$,

$$a' = \{P' \in \mathcal{P} \mid \forall (x', y') \in sp(\gamma(a), r). P'(x', y')\}$$

Continuing the Derivation of sp

$$a' = \{P' \in \mathcal{P} \mid \forall (x', y'). (x', y') \in sp(\gamma(a), r) \rightarrow P'(x', y')\}$$

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Then $(x', y') \in sp(\gamma(a), r)$ means

$$\exists x, y. (x, y) \in \gamma(a) \wedge R(x, y, x', y')$$

which, after expanding γ , gives

$$\exists x, y. \left(\bigwedge_{P \in a} P(x, y) \right) \wedge R(x, y, x', y')$$

We then plug this expression back into a' definition. Because the existentials are left of implication, the result is:

$$a' = \{P' \in \mathcal{P} \mid \forall x, y, x', y'. \left(\bigwedge_{P \in a} P(x, y) \right) \wedge R(x, y, x', y') \rightarrow P'(x', y')\}$$

Example of Analysis Result

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At all other points: $\perp$, which is:

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We remove predicates that do not hold